



ESOC-X-HACK CODE-BROWN

DevBytes

THE PROBLEM STATEMENT

Unsafe, Ignored, Pothole-ridden city roads

- Causes accidents and vehicle damage, especially for two-wheelers.

Inefficient complaint systems

- Manual processes hinder timely resolutions and reduce transparency.
- No status updates available for community-submitted complaints.

Duplicate or unverified reports

- Flood the system, leading to confusion and wasted resources which is one of the main reason behind delay.



SOLUTION ?



e-NIVARAN

SMART POTHOLE DETECTION & CIVIC MANAGEMENT

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What is eNivaran?



ROAD FIXING PRIORITY

- Identifies potholes using deep learning models from uploaded images and tells its priority.



SMART CIVIC REPORTING

- Enables users to file, track, and upvote complaints through a web interface.



ADMIN MANAGEMENT SYSTEM

- Admins can verify, manage, upload updates and resolve reports on with authority.



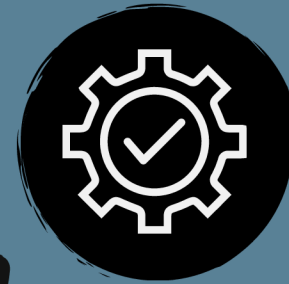
DUPLICATE CHECK SYSTEM

- Uses K-Means clustering and XGBoost to prevent duplicate complaints based on text, location, and image similarity.



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Key Features



ML-Based Pothole Detection

- Identifies potholes from images using trained models.



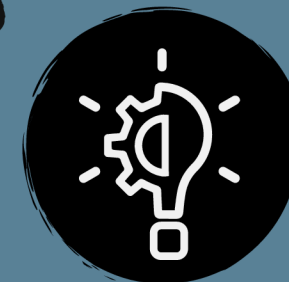
ML-Powered Duplicate Filtering

- Uses KMeans and XGBoost to avoid redundant complaints.



Role-Based Dashboards

- Separate user and admin views for seamless management.



Live Complaint Tracking

- Real-time status updates with upvotes and remarks from admin.

ML POTHOLE DETECTION



Advanced ONNX & YOLO Integration

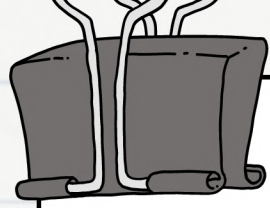
- Leverages a co-lab our trained ONNX model with YOLO architecture to deliver precise pothole detection.

Robust Image Processing Pipeline

- Utilizes OpenCV, PyTorch, and Pillow for dynamic image manipulation and comprehensive analysis.

Automated Detection Output Generation

- Produces both annotated detection images and detailed JSON results for seamless reporting.



PREVENTING REDUNDANCY



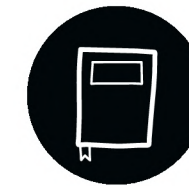
Text Similarity Analysis

- Combines Sentence-BERT with TF-IDF and Cosine Similarity.



Visual Similarity Detection

- Utilizes ResNet50 features for image matching.



Location Proximity Check

- Leverages Geopy for determining nearby locations.



Decision Fusion

- Applies an XGBoost classifier for final classification.



Duplicate complaint detected. Similar to complaint ID 2.



Flask Web Application

"Difficult than what it seems?
That's like finding a pothole on a
smooth road—just when you
think it's all clear, you hit a bump
and laugh it off!"



Seamless User Access

Secure authentication with persistent sessions keeps users logged in safely.



Smart Detection Tools

Dynamic pages enable users to submit and review road issue reports intuitively.



Streamlined Complaint Handling

Dedicated interface integrates AI modules to analyze and detect potholes effectively.



Comprehensive Admin Oversight

Dashboard provided for real-time report management and system monitoring.

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SYSTEM ARCHITECTURE

01

Complaint Intake:

Users upload images/complaints to initiate pothole detection.

02

ML Detection Engine:

The AI model processes submissions to identify potholes.

03

Duplication Guard:

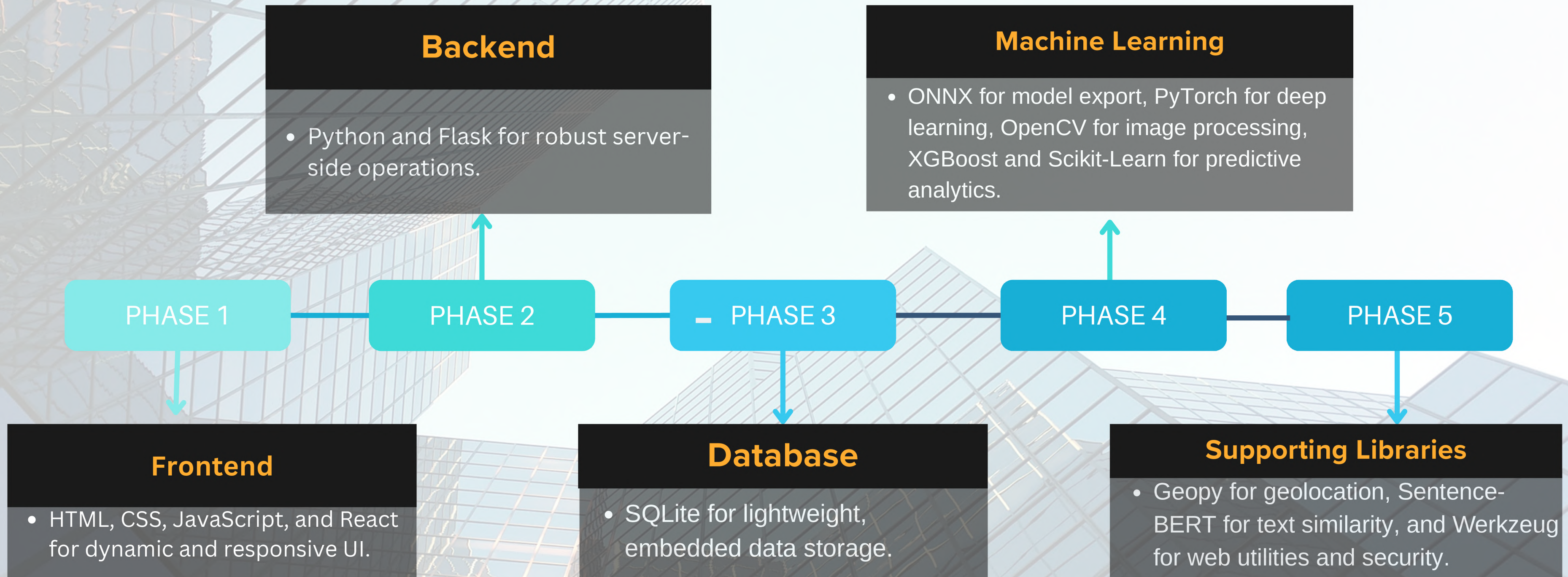
A duplication module filters out redundant entries.

04

InSQLite Data Hub:

Data is stored in SQLite databases and made visible to admins and users.

Tech Stack Implementation Timeline





THANK YOU

Feel free to ask any questions you may like.